7 A

PROGRAM To Implement a Decision Tree AND to DISPLAY IT

Number of Training samples

105

Number of Testing samples

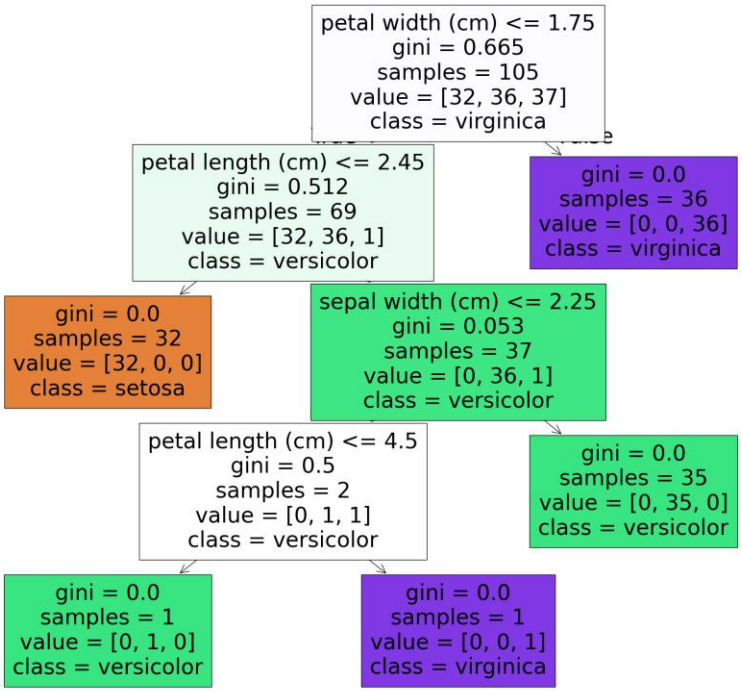
45

Class of Testing Samples

[0 2 2 1 2 0 1 0 1 0 2 0 0 1 0 2 2 0 0 1 1 0 2 1 0 0 1 1 1 2 0 1 1 1 0 2 1

 1 0 1 0 2 1 0 0]

```
|--- feature_3 <= 1.75
|   |--- feature_2 <= 2.45
|   |   |--- class: 0
|   |--- feature_2 >  2.45
|   |   |--- feature_1 <= 2.25
|   |   |   |--- feature_2 <= 4.50
|   |   |   |   |--- class: 1
|   |   |   |--- feature_2 >  4.50
|   |   |   |   |--- class: 2
|   |   |--- feature_1 >  2.25
|   |   |   |--- class: 1
|--- feature_3 >  1.75
|   |--- class: 2
```



7 B

PROGRAM To Calculate Accuracy of Decision Tree

Number of Training samples

105

Number of Testing samples

45

Class of Testing Samples

[2 1 0 0 0 1 2 2 0 1 0 1 1 1 2 0 2 2 2 2 2 2 0 2 1 1 0 1 0 2 2 0 1 1 1 0 2

0 2 1 0 0 1 0 1]

Accuracy

0.9555555555555556

7C

Training Samples

```
     Gender      Height
379       1  167.812868
382       1  151.547080
632       2  182.916158
581       2  175.775977
361       1  172.406437
..      ...        ...
466       2  177.008856
720       2  180.900459
758       2  174.467375
665       2  186.123714
721       2  182.624688

[560 rows x 2 columns]
```

Testing Samples

```
     Gender      Height
508       2  173.688749
740       2  187.507311
673       2  191.408458
405       2  166.325644
603       2  190.687835
..      ...        ...
730       2  186.290447
745       2  193.459436
570       2  180.618532
658       2  186.966280
208       1  159.593768

[240 rows x 2 columns]
```

criterion=absolute_error:

```
|--- Gender <= 1.5
|   |--- value: [160.8]
|--- Gender >  1.5
|   |--- value: [179.6]
```

Program Executed successfully